

can't even pick up a broom! But the next generation that is being created—in the shapes of animals—could well be more capable. There is a good chance that an Aibo could be sitting at your door, at guard in the night after a hard day's work inside the house.

Great Expectations

As humans, we always have great expectations from our friends, families and ourselves. But will our robot friend understand it? To be honest, it's best if they didn't. So long as they just did what is asked of them, we'll be happy!

And no, this is not fantasy. Already, some toy-like robots are reminding the elderly to take medicine and keep appointments. And future plans call for bundling more advanced mechanisms into roving maids and butlers.

Ah... a butler that will be at hand for your whims and fancies. While most inventors maintain that robots are 'almost intelligent', the question is, will there be a time when they could be more intelligent than us? And this question wakes the doomsday soothsayers from their slumber. But they can rest easy for a while as inventors and creators of the modern day maids and butlers are still worrying about issues like movement and response times.

Hollywood has already done its bit to scare us out of our wits by depicting robots as evil beings whose only purpose of existence is to take over the earth. One question: why? What on earth might push them to take over? Is it the fact that they no longer want to be ruled and want to be masters of their own destiny? But that would mean they are able to think and understand emotions and have aspirations. Unfortunately, that's happening only in Hollywood.

When Can You Buy One?

Your dreams of having a butler and/or a maid scooting around the house may not be fulfilled this decade, but some of the early developed bots are functioning as guards, delivering

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hospital food trays and carrying documents from one office to another.

The Japan Robot Association (JRA) had estimated that by 2002, nearly 11,000 service robots would have been deployed, 65 per cent of them in hospitals and nursing homes. JRA projected that by 2005, healthcare robots would be a \$250 million market, which would then hit \$1 billion by 2010—approaching 10 per cent of the total market that year. As for personal bots, a panel of industry and academic experts predicted they will be as common as PCs and cell phones within 10 to 15 years. You could finally draw a bath that's exactly 29.4° C every day!

Growing Up A Bot

Aibo, Asimo, Robosapien and any of the other similar bots have one thing missing when compared to Rosie. They do not have a character of their own.

Bots like Aibo can learn tricks and respond to voice commands, and are touted to be the next big thing, but what good are a few tricks when you want the whole circus?

While researchers everywhere are fascinated by the potential of home robots and cyber-companions, there are labs that are counting on biological approaches to help produce mechanical beings—known as evolutionary robotics.

The basic idea of evolutionary robotics is to be able to raise a machine like a child, letting it learn from its own experiences and sensory impressions, rather than feeding it canned software written by humans. Leading places for such research include the University of Sussex in Britain, Switzerland's University of Lausanne, and the MIT and Michigan State University in the US.

MIT's Cog and Kismet are probably the most famous of the self-educated bots. But no matter how personable the Cog may seem, a robot with human-level intelligence still remains a faraway dream.

The Evolution Chart

Researchers in corporate and academic labs are racing to develop working models that, within a few years, may become our cohabitants and co-workers. The leaders include companies such as Matsushita Electric Industrial, NEC, Sony and Omron and are literally investing millions in the development of personal robots. Here are some bots that have so far seen light of they day:

Asimo (Honda Motors): One of the first 'robot sapiens' to be developed was Honda's Asimo, a 1.2-metre-tall android that resembled a child astronaut. It could stride like a human, climb stairs, and negotiate corners. It could even turn the lights on and off, and to show off, it could walk a figure eight!

Asimo, however, was blind, deaf and dumb—to be able to perform its antics, it had to be remotely controlled. It may seem that a remote-controlled robot is no big deal, but like we've mentioned, the very fact that it could stride and climb stairs is a massive feat of engineering and computer science.



Aibo (Sony Electronics): Aibo is controlled by its Mind 2 Software located on a removable Memory Stick. This Memory Stick dictates Aibo's behaviour. This is even possible using a PC or a mobile device.

In daily life, Aibo can communicate with and entertain its owner. As time passes, Aibo also starts to identify its owner's voice and face and it can even recharge itself!



Roomba (iRobot): Developed by iRobot, Roomba is an intelligent robotic vacuum cleaner. All Roomba 'Robotic Floorvacs' feature iRobot's AWARE Robotic Intelligence System, which uses lots of sensors to monitor Roomba's environment, and adjusts the bot's behaviour up to 67 times per second, ensuring that Roomba cleans effectively, intelligently and safely.

Roomba is also the first cleaning robot to have hit the market commercially, and has already tasted success.

